Deepam Ameria

Availability – May 2026 | dameria@cs.cmu.edu | LinkedIn | +1 (412) 918-9360 | Website

Education

Carnegie Mellon University – Robotics Institute, School of Computer Science — Aug. 2024 – May 2026
Master of Science in Robotic Systems Development | GPA: 4.20/4.0 — Pittsburgh, PA
Coursework: Deep Learning; Robot Learning; Optimal Control and Reinforcement Learning; Computer Vision; Manipulation, Estimation, and Control, Robot Autonomy. *Teaching Assistant:* Mechatronic Design. Medical Robotics

University of Mumbai – K.J. Somaiya College of Engineering — Aug. 2019 – May 2023
Bachelor of Technology in Mechanical Engineering | GPA 9.08/10.0 — Mumbai, India
Coursework: Robotics and AI; Deep Learning in Autonomous Vehicles; Programming in C; Mechatronics, Electric Vehicles

Experience

Robotics Software Intern, *AlphaZ, Inc.,* *Los Angeles, California* — Jun. 2025 – Aug. 2025

- Delivered an autonomous stair-climbing module for the legged robot platforms Unitree B2 and Unitree Go2, enabling reliable multi-floor navigation and patrolling.
- Implemented deep learning-based stair perception with depth fusion and real-time lateral error estimation, with a PID closed-loop controller for precise alignment and ascent.
- Deployed the Fixposition GNSS on the Clearpath Husky A300, providing precise position for outdoor patrolling.

Graduate Engineer R&D, *Yulu Bikes Pvt. Ltd.,* *Bangalore, India* — Jul. 2023 – Feb. 2024

- Implemented a real-time state-estimation pipeline for bike orientation using CAN and IMU decoding with Kalman filtering. Validated on 200+ runs, delivering stable bike-tilt signals for control and analytics.
- Trained and deployed a lightweight time-series ML model on fused IMU data to predict fall risk during evasive maneuvers, with low-latency inference integrated into the ride software stack.

Self-Driving and Powertrain Engineer, *Orion Racing India,* *Mumbai, India* — Jan. 2022 – May 2023

- Integrated a SICK MRS1000 LiDAR and built a real-time point-cloud pipeline, applying DBSCAN clustering to segment objects for mapping and navigation.
- Implemented and tuned a pure pursuit lateral controller for reference tracking and lane keeping.
- Led design of a 7.9 kWh Li-Po battery pack, integrating BMS telemetry and fault reporting into the vehicle software stack.

Projects

Autonomous Lunar Rover for terrain manipulation (Advisor: Dr. William "Red" Whittaker) — Sep. 2024 – Present

- Building an autonomous terrain-modification lunar rover using ROS 2 and C++ with a linear-actuated dozer arm, enabling both autonomous and teleoperated crater grooming for craters up to 1m in diameter.
- Designing a perception and planning stack leveraging high-resolution RGB-D data from a ZED 2i camera and deep learning methods to detect craters, plan tool operations, and provide feedback for system validation.
- Containerized the ZED SDK on NVIDIA Jetson Orin to deploy in Docker, enabling active mapping of simulated lunar terrain with portable, scalable integration.

Motion Tracking with Real-Time Defense Mechanism, *Carnegie Mellon University* — Feb. 2025 – May 2025

- Integrated depth camera and interfaced Franka Arm manipulator using ROS to perceive and track motions and improve human-robot interaction.
- Leveraged YOLOv8-seg for object detection and segmentation, and DeepSORT for motion tracking.
- Exploring imitation learning to map attacker motion to real-time defensive actions.

Optimal Model-Based Control for Humanoid Soccer, *Carnegie Mellon University* — Feb. 2025 – May 2025

- Developed a simulated soccer-kicking framework for the Unitree G1 humanoid to achieve stable ball strikes toward a target.
- Formulated a QP to compute foot impulse and contact position, with trajectories refined via Direct Collocation.
- Implemented Infinite Horizon Linear Quadratic Controller (LQR) for balance before/after impact and Time-Varying LQR for accurate trajectory tracking during the kick.

Multi-Faceted Computer Vision, *Carnegie Mellon University* — Aug. 2024 – Dec. 2024

- Implemented Spatial Pyramid Matching for scene classification using feature extraction.
- Performed 3D Reconstruction using structure-from-motion, multi-view stereo, and Photometric Stereo to recover surface normals and reconstruct 3D shapes from varying lighting conditions.
- Applied FasterRCNN for object detection and a simple IoU-based multi-object tracker to detect and track objects across frames.

Skills

- **Programming and Platforms:** Python, C++, MATLAB, Embedded C++, ROS2, Linux, Mujoco, SolidWorks, ANSYS, EAGLE
- **Software Development and Libraries:** PyTorch, OpenCV, Bash, Git, Docker, AWS, Data Structures and Algorithms
- **Robotics:** Kinematics and Dynamics, Deep Learning, Reinforcement Learning, Control Systems, Sensor Fusion and Calibration